

Chap 6

Techniques of Integration

§1 Integration by Parts

* Integration by Parts

$$\frac{d}{dx} [f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$$

$$\Rightarrow f(x)g(x) = \int f'(x)g(x)dx + \int f(x)g'(x)dx$$

$$\Rightarrow \int f(x)g'(x)dx = f(x)g(x) - \int f'(x)g(x)dx$$

This is called the formula for integration by parts.

$$\cdot \int f(x)dx = xf(x) - \int xf'(x)dx$$

Exam 1 Find $\int x \sin x dx$.

$$\begin{aligned} \text{(sol)} \quad \int x \sin x dx &= x(-\cos x) - \int (-\cos x) dx \\ &= -x \cos x + \int \cos x dx \\ &= -x \cos x + \sin x + C \end{aligned}$$

Exam 2 Evaluate $\int \ln x dx$.

$$\begin{aligned} \text{(sol)} \quad \int \ln x dx &= x \ln x - \int x \frac{1}{x} dx \\ &= x \ln x - \int dx \\ &= x \ln x - x + C \end{aligned}$$

Exam 3 Find $\int t^2 e^t dt$.

$$\begin{aligned} \text{(sol)} \quad \int t^2 e^t dt &= t^2 e^t - \int 2te^t dt \\ &= t^2 e^t - \left[2te^t - \int 2e^t dt \right] \\ &= t^2 e^t + 2te^t + 2 \int e^t dt \\ &= t^2 e^t + 2te^t + 2e^t + C \\ &= (t^2 + 2t + 2)e^t + C \end{aligned}$$

Exam 4 Evaluate $\int e^x \sin x \, dx$.

$$\begin{aligned} \text{(sol)} \quad \int e^x \sin x \, dx &= e^x(-\cos x) - \int e^x(-\cos x) \, dx \\ &= -e^x \cos x + \int e^x \cos x \, dx \\ &= -e^x \cos x + e^x \sin x - \int e^x \sin x \, dx \end{aligned}$$

$$\Rightarrow 2 \int e^x \sin x \, dx = e^x(\sin x - \cos x) + 2C$$

$$\therefore \int e^x \sin x \, dx = \frac{e^x}{2}(\sin x - \cos x) + C$$

$$* \quad \int_a^b f(x) g'(x) \, dx = f(x) g(x) \Big|_a^b - \int_a^b f'(x) g(x) \, dx.$$

Exam 5 Calculate $\int_0^1 \tan^{-1} x \, dx$.

$$\begin{aligned} \text{(sol)} \quad \int_0^1 \tan^{-1} x \, dx &= x \tan^{-1} x \Big|_0^1 - \int_0^1 \frac{x}{1+x^2} \, dx \\ &= \tan^{-1} 1 - \frac{1}{2} \int_0^1 \frac{2x}{1+x^2} \, dx \\ &= \frac{\pi}{4} - \frac{1}{2} \ln(1+x^2) \Big|_0^1 \\ &= \frac{\pi}{4} - \frac{1}{2} \ln 2 \end{aligned}$$

Exam 6 Prove the reduction formula

$$\int \sin^n x \, dx = -\frac{1}{n} \cos x \sin^{n-1} x + \frac{n-1}{n} \int \sin^{n-2} x \, dx \quad \dots \quad (*)$$

where $n \geq 2$ is an integer.

$$\begin{aligned} \text{(sol)} \quad \int \sin^n x \, dx &= \int \sin^{n-1} x \sin x \, dx \\ &= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x \cos^2 x \, dx \\ &= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x (1 - \sin^2 x) \, dx \\ &= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x \, dx - (n-1) \int \sin^n x \, dx \\ \Rightarrow n \int \sin^n x \, dx &= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x \, dx \\ \Rightarrow \int \sin^n x \, dx &= -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} \int \sin^{n-2} x \, dx \end{aligned}$$

$$\begin{aligned}
(\text{Exam}) \quad \int \sin^5 x \, dx &= -\frac{1}{5} \sin^4 x \cos x + \frac{4}{5} \int \sin^3 x \, dx \\
&= -\frac{1}{5} \sin^4 x \cos x + \frac{4}{5} \left[-\frac{1}{3} \sin^2 x \cos x + \frac{2}{3} \int \sin x \, dx \right] \\
&= -\frac{1}{5} \sin^4 x \cos x - \frac{4}{15} \sin^2 x \cos x - \frac{8}{15} \cos x + C
\end{aligned}$$

Exercise 48

$$\int \cos^n x \, dx = \frac{1}{n} \sin x \cos^{n-1} x + \frac{n-1}{n} \int \cos^{n-2} x \, dx \quad \dots \quad (**)$$

$$\begin{aligned}
(\text{Exam}) \quad \int \cos^4 x \, dx &= \frac{1}{4} \cos^3 x \sin x + \frac{3}{4} \int \cos^2 x \, dx \\
&= \frac{1}{4} \cos^3 x \sin x + \frac{3}{4} \left[\frac{1}{2} \cos x \sin x + \frac{1}{2} \int dx \right] \\
&= \frac{1}{4} \cos^3 x \sin x + \frac{3}{8} \cos x \sin x + \frac{3}{8} x + C
\end{aligned}$$

* From (*), we have

$$\begin{aligned}
&\frac{n-1}{n} \int \sin^{n-2} x \, dx = \frac{1}{n} \cos x \sin^{n-1} x + \int \sin^n x \, dx \\
\Rightarrow \int \sin^{n-2} x \, dx &= \frac{1}{n-1} \cos x \sin^{n-1} x + \frac{n}{n-1} \int \sin^n x \, dx
\end{aligned}$$

Let $m = n - 2$.

$$\Rightarrow \int \sin^m x \, dx = \frac{1}{m+1} \cos x \sin^{m+1} x + \frac{m+2}{m+1} \int \sin^{m+2} x \, dx$$

From (**), we have

$$\int \cos^m x \, dx = -\frac{1}{m+1} \cos^{m+1} x \sin x + \frac{m+2}{m+1} \int \cos^{m+2} x \, dx$$